

Large-Scale Convex Optimization

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1 Introduction

Large Scale Problem of dimension n but iterations $\ll n$ desired

Convex One of the only problem classes that are "solvable"

Mathematical Optimization

$$\begin{aligned} & \text{minimize } f(x) \\ & \text{s.t. } g_i(x) \leq 0, \quad i = 1, \dots, n_g \\ & \quad h_i(x) = 0, \quad i = 1, \dots, n_h \end{aligned} \quad (1)$$

- $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ decision variable

(most of our algorithms also work for $n \rightarrow \infty$)

- f objective function

- $C = \{\xi \in \mathbb{R}^n : g(\xi) \leq 0, h(\xi) = 0\}$ feasible set

1.1 Important Definitions

- x^* is a *global minimum* if $f(x^*) \leq f(x)$

- x^* is a *local minimum* if there exists $\epsilon > 0$ s.t.

$$f(x^*) \leq f(x) \quad \forall x \in C \cap B_\epsilon(x^*)$$

$B_\epsilon(x^*) := \{\xi \in \mathbb{R}^n : |\xi - x^*| < \epsilon\}$ open ball, center x^* , radius ϵ

1.2 Existence of minimum

Counter examples

a) unbounded level sets, f.e. $1/x$

b) C open f.e. $(0, 1)$ but minimum at f.e. 0

c) f not l.s.c. (lower semi-continuous)

Proposition 1. f (lower-semi-)continuous, $f(x) \rightarrow \infty$ for $|x| \rightarrow \infty$, C closed $\Rightarrow \exists$ minimizer of (4) described by: $\min_{x \in C} f(x)$ and $\operatorname{argmin}_{x \in C} f(x)$

Examples

x :

- assets in a portfolio

- control inputs

- schedule assignment

- resource allocation

C :

- all possible trade assets

- actuation limits

f :

- cost (negative returns)

- deviation from target

- waiting times / delays

- risk (a certain resource fails)

First Order Algorithmus

Initialize x_0

for $k = 0, \dots, \# \text{iterations} - 1$

$(f(x_k), \nabla f(x_k)) \leftarrow$ call first-order oracle

Determine x_{k+1} based on ∇f .

end

Analytic, arithmetic complexity

Definition 1 (Lipschitz continuity). $q : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is Lipschitz with constant L if: $|q(x) - q(y)| \leq L|x - y| \forall x, y \in \mathbb{R}^n$

Class of OP P with $C = [0, 1]^n$ and f is L^∞ -Lipschitz

Proposition 2. For any algorithm $\exists$ problem in P , s.t. achieving $|f(x_N) - f(x^*)| < \epsilon$ requires

$$N \geq \left(\left\lceil \frac{L}{2\epsilon} \right\rceil \right)^n - 1$$

Example

(for $L=1, \epsilon = 0.0005, n=27, N$ larger than #atoms in universe)

Proof. **Idea** Construct f where $(f(x_0) = 0, \nabla f(x_0) = 0)$, $(f(x_1) = 0, \nabla f(x_1) = 0), \dots$ but the actual $\min_{x \in C} f(x)$ is small.

Grid(x1,x2)

raster $1/3$, 9 boxes in $(1,1)$, for $N \leq 7$ (8 steps) one grid cell is not visited

Hence $f(x_i) = 0, i \in [0, 7]$ but $f(x^*) = -L/6$

Generalization

- Partition unit cube into s^n small boxes with side length $1/s$ and $\min_{x \in C} f(x) = -L/2s$ - therefore $f(x_i) - f(x_{star}) \geq L/2s$ for $i = 0 \dots s^n - 2$ - roughly ... - therefore $N = \dots$

□

Definition 2. The optimization problem 4 is convex if f and g_i are convex functions, $i = 1, \dots, n_g$, and h is affine.

Definition 3. Function $q : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex (affine) if for any $x, y \in \mathbb{R}^n$

$$q(\theta x + (1-\theta)y) \leq \theta q(x) + (1-\theta)q(y) \quad \forall \theta \in [0, 1]$$

Software Frameworks

- CVX Python - Yalmip

Proposition 3. x^* local minimum of (4), if (4) convex, then x^* global minimum of (4)

Proof. Counter example, $\exists y \neq x^* \in C$ such that $f(y) \leq f(x^*)$ □

1.3 Recitation

LOOK AT SLIDES or FIND r1.md

2 Convex sets and convex functions

Definition 4 (Convex Set). A set C is convex if and only if $\forall x, y \in C$ and $\forall \theta \in [0, 1]$: $\theta x + (1-\theta)y \in C$.

Examples of convex sets:

- hyperplane $\{x \in \mathbb{R}^n \mid a^T x = b\}$
- half-space $\{x \in \mathbb{R}^n \mid a^T x \leq b\}$
- polyhedron $\{x \in \mathbb{R}^n \mid Ax \preceq b, Cx = d\}$
 $A \in \mathbb{R}^{q \times n}, C \in \mathbb{R}^{r \times n}, b \in \mathbb{R}^q, d \in \mathbb{R}^r$
- ...more...

2.1 Operations that preserve convexity (sets)

- **Intersection** C_1, C_2 convex $\Rightarrow C_1 \cap C_2$ convex
- **Image under affine map** $C \subseteq \mathbb{R}^n$ convex $\Rightarrow \{Ax + b \mid x \in C\}$ convex
- inverse image of an affine map: ...

2.2 Separating Hyperplane Theorem

Theorem 1. $C \subseteq \mathbb{R}^n$ non-empty closed convex set, $y \notin C$
 $\Rightarrow \exists a \neq 0, b \in \mathbb{R}$ s.t. $a^T x + b < a^T y + b, \forall x \in C$

Proof. **Claim** $\exists \hat{x} \in C$ s.t. $|\hat{x} - y| \leq |x - y| \quad \forall x \in C$

Proof of claim $|x - y|$ has bounded level sets, C is non-empty and closed
 $\Rightarrow \exists \hat{x} \in \operatorname{argmin}_{x \in C} |x - y|$

Hyperplane, we choose $a := y - \hat{x}, b := -a^T \hat{x} = -(y - \hat{x})^T \hat{x}$

As a result, $a^T x + b = (y - \hat{x})^T (x - \hat{x})$ and therefore $a^T y + b = |y - \hat{x}|^2 > 0$. The following claim shows that the hyperplane $a^T y + b$ separates C and y .

Claim $a^T y + b \leq 0 \quad \forall x \in C$

Proof of claim Assume not. $\rightarrow \exists x \in C$ s.t. $(y - \hat{x})^T (x - \hat{x}) > 0$

PARAMETRIZE θ

Contradiction $\hat{x}$ nearest point to y

(Details in Lecture notes) □

Corollary 1. A closed convex set $C \subseteq \mathbb{R}^n$ is the intersection of the closed half-spaces that contain C .

Proof. S intersection of closed half-spaces that contain C

1) $C \subseteq S : x \in C \Rightarrow x$ is contained in every half-spaces that contains C
 $\Rightarrow x$ is also contained in the intersections of half-spaces that contains C
 $\Rightarrow x \in S$

2) $S \subseteq C : \text{Assume not} \rightarrow \exists \hat{x} \in S$ with $\hat{x} \notin C$. By the Separating Hyperplane Theorem there exists a hyperplane that separates $\hat{x}$ from C . That means there exists a closed half-space that contains C but not $\hat{x}$, hence $\hat{x} \notin S$, contradiction. □

2.3 Support function

Idea represent any closed convex set by its supporting hyperplanes

Support Function: $\sigma_C(a) = \sup_{x \in C} a^T x$

CALCULATION EXAMPLE

If we know the $\sigma_C(a)$, we arrive at

$$\begin{aligned} C &= \bigcap_{a \in \mathbb{R}^n} \{x \in \mathbb{R}^n \mid a^T x - \sigma_C(a) \leq 0\} \\ &= \{x \in \mathbb{R}^n \mid \sup_{a \in \mathbb{R}^n} a^T x - \sigma_C(a) \leq 0\} \end{aligned}$$

Definition 5. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if its epigraph is a convex set, where

$$\operatorname{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}$$

$\rightarrow$ this provides a link between convex sets and functions

2.4 Operations that preserve convexity (functions)

- the pointwise maximum of convex functions is convex
- the sum of convex functions is convex
- $f(Ax + b)$ is convex if f is convex

How to check if f is convex?

- if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable, $\partial^2 f / \partial x^2 \succeq 0 \quad \forall x \in \mathbb{R}^n$
- if $g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(t) = f(x + tv)$ convex in $t \quad \forall x, v \in \mathbb{R}^n$, then f is convex
- composition of simple convex function with convexity preserving operations

Extended real numbers $\bar{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$

Indicator function $\psi_C(x) := \begin{cases} +\infty & \text{if } x \notin C \\ 0 & \text{if } x \in C \end{cases}$

$\rightarrow$ this provides another link between convex sets and functions

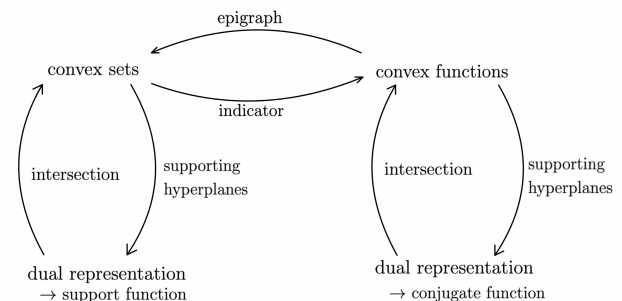
We can write $\min_{x \in C} f(x)$ as $\min_{x \in \mathbb{R}^n} f(x) + \psi_C(x)$

Definition 6 (3). $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is called proper if f is bounded below and if $\exists x \in \mathbb{R}^n$ s.t. $f(x) < \infty$

Definition 7 (Legendre Transformation). The conjugate function of $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is defined as $f^*(y) = \sup_{x \in \mathbb{R}^n} y^T x - f(x)$

IMAGE F-STAR

2.5 Summary of Concepts



Theorem 2 (THEOREM2?).

2.6 Recitation

Convex Sets

A set C is convex if and only if for all $x, y \in C$ and $\theta \in [0, 1]$:

$$\theta x + (1 - \theta)y \in C$$

Convex Cone

conic combination

Given $x_1, \dots, x_n$

any point of the form:

$$\theta_1 x_1, \dots, \theta_n x_n$$

$$\theta_i \geq 0$$

convex cone

XXX

Positive Semidefinite Cone

Notation

$\mathbb{S}^n$ set of symmetric $n \times n$ matrices

$\mathbb{S}_+^n$ HHH

$\mathbb{S}_{++}^n$ HHH not convex cone

Example

Sylvester Condition

Convex Functions

Definition

Methods for establishing convexity

1. Verify from definition
2. Second order condition
3. Operations that preserve convexity

Log-Sum-Exp

$$f(x) = \log(e_1^x + \dots + e_n^x)$$

differentiable approximation of $\max(x)$

How to check convexity?

Second-order condition $\nabla^2 f \geq 0$

Nonnegative Weighted Sum

$\alpha(f_1 + f_2)$ convex if f_1, f_2 convex, $\alpha > 0$

$f_1, \dots, f_m$ convex, $w_1, \dots, w_m \geq 0 \Rightarrow w_1 f_1 + \dots + w_m f_m$ convex

Composition with Affine Function

$$g(x) = f(Ax + b)$$

Examples

Log barrier for linear inequalities $\rightarrow$ transforms constrained problem in unconstrained

Norm Function

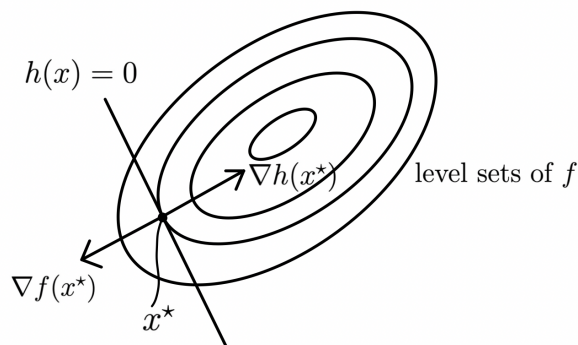
Composition

$$f(x) = h(g(x))$$

3 KKT and Lagrange Duality

3.1 Example

Optimization problem: $\min_{x \in \mathbb{R}^2} f(x)$ s.t. $h(x) = 0$



We note the following: $\nabla f(x^*)$ and $\nabla h(x^*)$ are colinear

$$\Leftrightarrow \exists \nu^* \in \mathbb{R} : \nabla f(x^*) + \nu^* \nabla h(x^*) = 0$$

$\Leftrightarrow f(x) + \nu^* h(x)$ is stationary at x^* , where ν^* can be interpreted as cost of violating constraint

3.2 Generalization

Generalization to $n \geq 2$ and presence of inequality constraints

$$f^* = \inf_{x \in \mathbb{R}^n} f(x) \text{ s.t. } h(x) = 0, g(x) \leq 0 \quad (2)$$

with corresponding Lagrange function

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^T g(x) + \nu^T h(x) \quad (3)$$

where $\lambda_i \geq 0, \nu_i \in \mathbb{R}$ are the dual variables or multipliers that can be interpreted as cost for violating constraints.

Proposition 4 (Weak Duality). The dual function

$$d(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, \nu) \text{ satisfies}$$

$$d(\lambda, \nu) \leq f^*, \forall \lambda \geq 0, \nu \in \mathbb{R}^{n_h}$$

Proof.

Definition 8 (Constraint qualification). C convex, Slater's Condition holds if $\exists \hat{x} \in \mathbb{R}^n$ s.t. $h(\hat{x}) = 0$ and $g(\hat{x}) < 0$

Proposition 5 (Strong Duality). If Slater's condition holds and (2) is convex then $\exists \lambda \geq 0, \nu \in \mathbb{R}^{n_h}$ s.t. $d(\lambda, \nu) = f^*$

Proof. GRAPHIC

3.3 KKT

Theorem 3 (KKT Conditions). Slater's condition holds and (2) is convex $\rightarrow x^* \in \mathbb{R}^n$ is a minimizer of the primal (2) and $(\lambda^* \geq 0, \nu^*) \in \mathbb{R}^{n_g} \times \mathbb{R}^{n_h}$ is a maximizer of the dual $\Leftrightarrow$

$$\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0 \quad \text{KKT-1 (Stationary Lagrangian)}$$

$$g(x^*) \leq 0, h(x^*) = 0 \quad \text{KKT-2 (primal feasibility)}$$

$$\lambda^* \leq 0, \nu^* \in \mathbb{R}^{n_h} \quad \text{KKT-3 (dual feasibility)}$$

$$\lambda^{*T} g(x^*) = 0 = \nu^{*T} h(x^*) \quad \text{KKT-4 (complementary slackness)}$$

In addition we have: $\sup_{\lambda \geq 0, \nu \in \mathbb{R}^{n_h}} q(\lambda, \nu) = \inf_{x \in C} f(x)$

QUESTION Proof?

Remark Without Slater, KKT 1 to 4 still implies x^* minimizes (2) and (λ, ν) maximizes the dual, but the converse is no longer true, there can be primal/dual minimizer/maximizer that do not satisfy KKT1-4

FORCE BALANCE

3.4 What if f, g not differentiable?

Example $\inf_{x \in \mathbb{R}^n} |Ax - b|^2 + |x|_1$

where (l_1) -norm not differentiable at 0

3.5 Subdifferential

for convex f...

Definition 9. $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ convex, the subdifferential of f at $\bar{x}$ is: $\partial f(\bar{x}) := \{\lambda \in \mathbb{R}^n \mid f(\bar{x}) \leq \lambda^T \bar{x} + \inf_{y \in \mathbb{R}^n} \{\lambda^T y - f(y)\}\}$

Proposition 6. $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex. $x^* \in \operatorname{argmin} \dots$

Proposition 7 (Relation to conjugate functions). f convex, $\operatorname{epi}(f)$ closed: $y \in \partial f(x) \Leftrightarrow x \in \delta f^*(y)$

3.6 Recitation 3

Information ML

Hard Margin SVM

Use hyperplane and support vectors for data classification.

SVM

Find the Maximum-Margin Hyperplane

Solve the Optimization Problem

- Introduce Lagrange multiplier $\alpha_i \geq 0$ for $i = 1, 2, \dots, N$
- ...
- ...
- Solve α^* by Strong Duality
- Obtain w^* and b^* using KKT

Soft Margin SVM

- Introduce some *slackness* ξ
- Point 2

Kernel Methods: Break the linearity

Introduce Nonlinear feature map $\phi(x) : \mathbb{R}^n \rightarrow \mathbb{R}^m$

Kernel $K(x_i, x_j) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$

4 Convex Optimization Problems

Recall general optimization Problem

$$\begin{aligned} & \text{minimize} && f(x) \\ & \text{s.t.} && g_i(x) \leq 0, \quad i = 1, \dots, n_g \\ & && h_i(x) = 0, \quad i = 1, \dots, n_h \end{aligned} \quad (4)$$

OPTIMAL VALUE

4.1 Feasibility Problem

$$\begin{aligned} & \text{minimize} && s \\ & \text{s.t.} && g_i(x) \leq s, \quad i = 1, \dots, n_g \\ & && h_i(x) = 0, \quad i = 1, \dots, n_h \end{aligned} \quad (5)$$

4.2 Linear Programming

$$\text{minimize } c^T x \quad \text{s.t. } Ax - b \geq 0, x \geq 0 \quad (6)$$

Derive dual problem:

$$\text{Step 1: } \mathcal{L}(x, \lambda_1, \lambda_2) = c^T x - \lambda_1^T (Ax - b) - \lambda_2^T x, \lambda_i \geq 0$$

$$\text{Step 2: } \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda_1, \lambda_2) = \begin{cases} \lambda_1^T b & \text{if } c - A^T \lambda_1 - \lambda_2 = 0 \\ -\infty & \text{if } c - A^T \lambda_1 - \lambda_2 \neq 0 \end{cases}$$

Step 3: Dual Problem (again linear program)

$$\text{maximize } b^T \lambda \quad \text{s.t. } c - A^T \lambda \geq 0, \lambda \geq 0 \quad (7)$$

Skech

- Polyhedron
- c-vector normal gives 'Levelsets'
- Optimal solution in or trough a corner (if exists)
- Proposition 8.** The optimal solution of a linear program (if it exists) lies always on the boundary of the feasible set and there exists an optimal solution that is a vertex of the feasible set.

Shortest Path

Analogie with Fluid

Soltuion greater 0, not optimal edges = 0

4.3 Quadratic Programming

$$\text{minimize } \frac{1}{2} x^T P x + q^T x \quad \text{s.t. } Gx \leq h, Ax = b$$

Interpretation: minimize a convex quadratic function over a polyhedron.

If $P = P^T$ is positive semi-definite then the problem is convex.

Example [optimal control] (basis for mpc)

Second-order cone program (SOCP)

minimize $f^T x$

$$\text{s.t. } |A_i x + b| \leq c_i^T x + d_i, Fx = g$$

Second-order cone $C_{n+1} = \{(x, t) \mid x \in \mathbb{R}^n, t \in \mathbb{R}, |x| \leq t\}$

Example [Markovitz portfolio optimization:]

- n number of assets/stocks
- x_i relative value of asset i
- p_i price change of stock i
- $p^T x$ overall return

Constraints

- $x^T \mathbf{1} = B$, total amount
- $x \geq 0$, no short position

CALCULATIONS

4.4 Semidefinite programming (SDP)

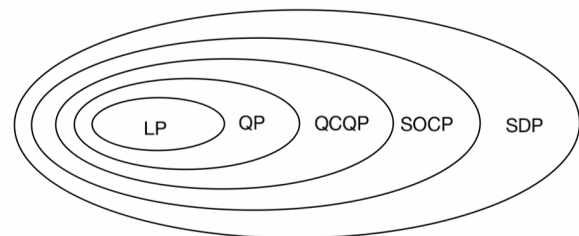
minimize $c^T x$

$$\text{s.t. } x_1 F_1, \dots, x_n F_n \preceq 0 \text{ and } Ax - b = b$$

→ the 'standard' form

$$\min_{x \in \mathbb{R}^{n \times n}} \text{tr}(CX)$$

Problem Classes



4.5 Recitation 4

Geometric Programming

Motivation

- Summary Change of variables, transformation of objectives and constraints

→ convex problem in standard form

- Monomial function

- Posynomial function

- Problem formulation

- Example

- Technique

Variable transformation $y_i = \log x_i$ on objective and constraints.

- Transformation

Sum of Squares

- Polynomial Optimization

→ f, g_i, h_i polynomials

General case intractable

- Nonnegative polynomials

Small adaption with γ

find largest γ such that $f(x) - \gamma$ nonnegative, NP Hard

→ chose γ very high, results in sum of squares

Definition A polynomial $f(x)$ is a sum of squares (SOS), if it can be written as

$$f(x) = \sum_i g_i^2(x) \quad g_i: \text{polynomial}$$

- Verification

$z(x)$ as vector that contains all polynomials of degree $\leq d$

Theorem 4 (SOS). $p(x)$ is an SOS if and only if $\exists Q$ such that $Q \succeq 0$ and $p(x) = z(x)^T Q z(x)$

Proof

Example

SOS for Lyapunov Stability Analysis

Dynamic

$$\dot{x}_1 = -x_1^3 + x_2$$

$$\dot{x}_2 = -x_1 - x_2$$

Equilibrium

$$x = (x_1, x_2) = (0, 0)$$

$$V(x) = ax_1^2 + bx_2^2 \quad \text{vdot} = \text{dVf}(x) = [2ax_1, 2bx_2]^T \text{dynvec}$$

verify $v_x > 0, \text{vdot} > 0$

5 Gradient methods - Part I

Definition 10 (smoothness). The function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth if $\nabla f(x)$ satisfies

$$|\nabla f(x) - \nabla f(y)| \leq L|x - y| \quad \forall x, y \in \mathbb{R}^n$$

This result (with Taylor's Theorem) in:

$$f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} |x - y|^2 \quad \forall x, y \in \mathbb{R}^n$$

Definition 11 (strong convexity). The function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is μ -strongly convex if it satisfies

$$f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2} |x - y|^2 \quad \forall x, y \in \mathbb{R}^n$$

5.1 Gradient Descent

Given x_0 and stepsize $T > 0$

$$x_{k+1} = x_k - T \nabla f(x_k) \quad \text{for } k = (k_0, \dots, k_N)$$

def gradient_descent(x_0, T):

for k in range(N):

x_next = x - T * df(x)

HERLEITUNG

Assume $f(x) = c_0 + b^T x + \frac{1}{2} x^T H x, H \succ 0 \Rightarrow H x^* = -b$

$$x_{k+1} - x^* = x_k - x^* - T(b + H x_k) = (I - TH)(x_k - x^*)$$

$$\begin{aligned} x_{k+1} - x^* &= x_k - x^* - T(b + H x_k) \\ &= (I - TH)(x_k - x^*) \end{aligned}$$

Convergence given by eigenvalue of $I - TH$, use $H = U \Lambda U^T$

$x_N - x^* = U(I - T\Lambda)^N U^T (x_0 - x^*) \rightarrow$ Convergence rate is dictated by $1 - T\lambda_i$

Optimal Step Size

$$\mu \leq h \leq L$$

$$T^* = \frac{2}{L + \mu}$$

GRAFIK

Convergence rate

$$\rho(T^*) = |1 - \frac{2L}{L + \mu}| = \frac{L - \mu}{L + \mu}$$

therefore with stepsize T^*

$$|x_N - x^*| \leq \epsilon \text{ if } N \geq \frac{\kappa+1}{2} \ln\left(\frac{|x_0 - x^*|}{\epsilon}\right)$$

5.2 Momentum-based methods

$$\begin{aligned} q_{k+1} &= q_k + T p_{k+1} \\ p_{k+1} &= (1 - 2dT) p_k - T \nabla f(q_k + \beta p_k) / L \end{aligned} \quad (8)$$

SPRING DAMPER ANALOGY

Nesterovs accelerated gradient methods

- for $T = 1, d = \frac{1}{\sqrt{k+1}}, \beta = \frac{\sqrt{k}-1}{\sqrt{k+1}}$

Heavy Ball (tuned quadratics)

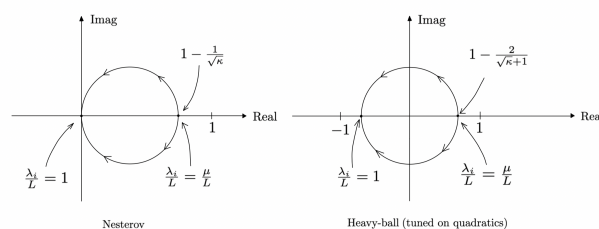
- for $T = \frac{2\sqrt{k}}{\sqrt{k+1}}, d = \frac{1}{\sqrt{k+1}}, \beta = 0$

What is the convergence rate?

EXAMPLE DIAGONALIZATION

EIGENVALUE analysis

ROOT Locus



- Nesterov on circle $c = (r/0), r = \lambda_i/L = \mu/L$

- Heavy ball circle $c = ((\lambda - L)/2, 0), r = \lambda + L$

$$C_{\text{Nesterov}} \left(1 - \frac{1}{\sqrt{\kappa}}\right)^N \approx \frac{|q_N - q^*|}{|q_0 - q^*|} \approx C_{\text{HeavyBall}} \left(1 - \frac{2}{\sqrt{\kappa+1}}\right)^N$$

Requires $N \geq 2\sqrt{\kappa} \ln\left(\frac{|q_0 - q^*|}{\epsilon}\right)$ to achieve $|x_N - x^*| \leq \epsilon$
Theorem 5. For any first-order method $\exists f: \mathbb{R}^\infty \rightarrow \mathbb{R}, \mu\text{-scv}, L\text{-sm}, \text{ s.t. } |x_k - x^*| \geq (1 - \frac{2}{\sqrt{\kappa+1}})^k |x_0 - x^*| \forall k \geq 0$

5.3 Recital 5 - More on Gradient Descent

Properties of Smooth Functions

- L-smoothness:

$$f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2} |x - y|^2 \quad \forall x, y \in \mathbb{R}^n$$

Gradient Descent

- Smooth and Convex

$x^{\star}$ argmin f

f is also L -smooth

select $\eta = \frac{1}{2L}$

summing up

- sufficient decrease

- this results in

$$f(x_T) \leq f(x_{T-1}) \leq \dots \leq f(x_1) \leq f(x_0)$$

- As a result, with stepsize $\eta = \frac{1}{2L}$, GD Converges with

$$f(x_T) - \min f(x) \leq \frac{2L}{T} \|x_0 - x^{\star}\|^2$$

- can do better, nestrov $1/T^2$

Properties of Strongly-Convex Functions

- μ -strong-convexity: $f(y) \geq \dots \mu/2 \dots$

...this implies

Smooth and Strongly-Convex

$\eta = \frac{1}{L}$ converges with ...

Stepsize

- guess if don't know L

- start with η ca ϵ

- double η until checkable condition does not hold

Line search

- 1-dimensional programming

- find η with optimization for every step

- can result in stepsize $\geq 1/L$ as it is for normal GD

Line search for Heavy Ball Method

- works also for quadratics

- conjugate GD, orthogonalize?

Adaptive Methods

- normalized GD

- AdaGrad-Norm (Adaptive Gradient estimation)

- AdaM (Adaptive Momentum estimation)

- AdamW

6 Gradient Methods - Part II

Projected gradient descent (smooth, strongly convex f)

Definition 12. $\text{prox}_C(x) = \arg\min_{y \in C} \frac{1}{2} \|x - y\|^2$

C closed convex

CAUCHY SCHWARZ

This implies: $|\text{prox}_C(x) - \text{prox}_C(y)| \leq \|x - y\|$

Proof Other Information. TODO

Algorithm

Proposition 9. $f: L$ -sm, μ -scv $\rightarrow$ projected GD with $T = \frac{2}{L+\mu}$ satisfies

$$\|x_N - x^{\star}\| \leq \|x_0 - x^{\star}\| \left(1 - \frac{2}{L+\mu}\right)^N \quad (\kappa \text{ still } \frac{L}{\mu})$$

Proof. Restricted on quadratic functions: $\frac{1}{2} x^T H x + b^T x + c$

□

when are projections computationally cheap?

- norm ball

- probability simplex

What if f is not strongly convex? ($\mu = 0$)

$\rightarrow$ idea: apply small amount of regularization

$f: \mathbb{R}^n \rightarrow \mathbb{R}$ L -smooth, convex

$$\hat{f}(x) = f(x) + \frac{\mu}{2} \|x - x_0\|^2$$

and

XXX (IEQ 1.2)

are satisfied $\forall x_0 \in \mathbb{R}^n, \mu > 0$, where $\dots x(\text{hat})^{\star}$ argmin $f(\text{hat})$

Proof. XXX

□

hence we can apply GD or Nesterov

calc

For Nesterov: $\dots e^{sqrt \frac{\mu}{L+\mu}} \dots$

sqrt ESSENCE of morning

chose $\dots \frac{2 \ln(N)}{N}$

BOX Hence if f smooth and (not strongly) convex we need approximately N tilde $L \|x^{\star} - x_0\|^2 / \epsilon$ iterations to reach $f(x_N) - f(x_0) \leq \epsilon$

What if f is non-smooth?

i.e. L_f Lipschitz but not necessarily differentiable

Example $f(x) = |x|$

Leads to oscillations with $\nabla f = \{+1 \mid -1\}$

Proposition 10 (Subgradient Method). Closed, convex set C contained in ball of $r = R$

Consider update rule: $x_{k+1} = \text{prox}_C(x_k - T g_k), \dots$ then $x_0, \dots$

Proof. NOT SHOWN

□

TABLE

GRAPH with rates, IMPORTANTe

6.1 Recitation

TODO

7 Stochastic gradient descent

Motivation Example

- Regression: $\tilde{y} = \phi(\tilde{x}_i; \theta) + \epsilon$

function approximator ϕ with parameter θ

- Data points: $(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$ with m large

- Minimized square loss: $\frac{1}{2m} \sum_{i=1}^m |\tilde{y}_i - \phi(\tilde{x}_i; \theta)|^2 \rightarrow \min$ with respect to (w.r.t.) θ

- Gradient: $-\frac{1}{m} \sum_{i=1}^m (\tilde{y}_i - \phi(\tilde{x}_i; \theta)) \frac{\partial}{\partial \theta} \phi(\tilde{x}_i; \theta)$

requires evaluating sum over m elements

$\rightarrow$ computationally intractable if m is large

Goal Obtain approximated solution quickly

$\Rightarrow$ Compute Stochastic Gradient

$$(\tilde{y}_i - \phi(\tilde{x}_i; \theta))^T \frac{\partial \phi}{\partial \theta} \Big|_{\tilde{x}_i; \theta}, i \sim \text{Unif}(\{1, \dots, m\})$$

$\Rightarrow$ this gradients are **unbiased**

More generally we consider the problem

$$\min_{x \in \mathbb{R}^n} F(x) = \min_{x \in \mathbb{R}^n} \mathbb{E}[f(x, \xi)]$$

Where ξ is a continuous or discrete Random Variable.

Recall definition of expected value:

$$\mathbb{E}_\xi[f(x, \xi)] = \begin{cases} \int_{\mathbb{R}^q} f(x, \xi) p_\xi(\xi) d\xi \\ \sum_{\xi} f(x, \xi) p_\xi(\xi) \end{cases}$$

ALGORITHM Stochastic gradient descent

Step 1: $\xi_k \leftarrow$ generate realization of ξ

Step 2: $x_{k+1} = x_k - T_k g(x_k, \xi_k)$, step size T_k

Stochastic gradient $g(x_k, \xi_k)$ examples:

$\nabla_x f(x, \xi)$, $\bar{\xi} \sim p_\xi$ or $\frac{1}{n_{mb}} \sum_{i=1}^{n_{mb}} \nabla_x f(x, \xi_i)$, $\xi_i \sim p_\xi$

$\Rightarrow$ The iterate x_k is now a random variable!

7.1 Assumptions on $F(x)$ and $g(x_k, \xi_k)$

A1 $F(x)$ is bounded below, ensures that $\min_x F(x)$ exists if F is L -smooth.

A2 $\mathbb{E}_\xi[g(x, \xi)] = \nabla F(x) \quad \forall x \in \mathbb{R}^n$ ensures that stochastic gradients are unbiased.

A3 $\exists M, M_v \geq 0$ s.t. $\text{Var}_\xi[g(x, \xi)] \leq M + M_v |\nabla F(x)|^2 \forall x \in \mathbb{R}^n$ ensures that variance is bounded.

Proposition 11. F μ -strongly convex and L -smooth. SGD with constant stepsize $0 < T < \frac{1}{L(M_v+1)}$ satisfies

$$\mathbb{E}[F(x_k)] - F(x^*) \leq \frac{TLM}{2\mu} + (1 - T\mu)^k (F(x_0) - F(x^*))$$

With stepsize $T = \frac{\ln(N)}{\mu N}$ we require about

$$N \sim \left(\frac{LM}{2\mu^2} + F(x_0) - F(x^*) \right) / \epsilon$$

iterations to ensure $\mathbb{E}[F(x_N)] - F(x^*) \leq \epsilon$

Proof with most important SGD-EQ. XXX

$$\mathbb{E}[F(x_{k+1}) | x_k] \leq F(x_k) - T |\nabla F(x_k)|^2 + XXX$$

(1)

Strong convexity implies:

$$F(x) \leq F(x^*) + \frac{1}{2\mu} |\nabla F(x)|^2 \quad \forall x \in \mathbb{R}^n$$

from there we can conclude: XXX

□

Remarks

$(1 - T\mu)^N \leq e^{-T\mu N}$ this in EQ

- $T_k = \frac{\ln(N)}{N}$ then $\mathbb{E}[XXX] \leq$

- $\sum_{k=0}^{\infty} T_k = \infty, \sum_{k=0}^{\infty} T_k^2 \leq \infty$

- $T_k = \frac{\beta}{\gamma+k}$

The role of mini batches

same analysis still holds if $M \rightarrow M/n_{mb}, M_v \rightarrow M_v/n_{mb}$

EQ

but we can also run SGD with step T/nmb and get same result

Advantage in computation if parallelization possible

Can we do non-(strongly-)convex functions?

Proposition 12. F, L -smooth, then SGD with stepsize $T \leq \frac{1}{L(1+M_v)}$ achieves

$$\mathbb{E} \left[\sum_{k=0}^{N-1} |\nabla F(x_k)|^2 \right] \leq N T L M + \frac{2}{T} (F(x_0) - F_{inf})$$

$F_{inf} = \inf_{x \in \mathbb{R}^n} F(x)$

Proof. similat to previous proposition, from (1) we infer:

$$\mathbb{E}[F(x_{k+1})] - \mathbb{E}[F(x_k)] \leq -\frac{T}{2} \mathbb{E}[X] X X$$

XXX

SUM

□

7.2 Table

7.3 Recitation

- SGD vs GD

- N is large, GD to costly

7.4 Methods to improve SGD

- Mini Batch

- Momentum, moving average of gradients

- Control Variates

- Variance Reduction Techniques

- SAGA stochastic averaging gradient

- Stochastic Variance Reduced Gradient (SVRG)

- Summary

Explanations on Code

8 Alternating Direction Method of Multipliers (ADMM)

Motivation

So far we looked at:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^m f_i(x) \quad (9)$$

where m is very large $\rightarrow$ SGD $\rightarrow$ comparably slow convergence

Now exploit parallelization

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^m f_i(x_i) \text{ s.t. } x_1 = x_2 = \dots = x_m$$

8.1 Dual ascent

Start with:

$$\min_{x \in \mathbb{R}^n} f(x) \text{ s.t. } Ax = b, A \in \mathbb{R}^{m \times n}$$

Derive dual:

$$\mathcal{L}(x, \lambda) = f(x) + \lambda^T Ax - \lambda^T b$$

$$\inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda) = - \underbrace{\sup_{x \in \mathbb{R}^n} \{(-\lambda^T A)x - f(x)\}}_{-f^*(-A^T \lambda)} - \lambda^T b$$

Hence the dual can be stated as follows:

$$\sup_{\lambda \in \mathbb{R}^m} \underbrace{-f^*(-A^T \lambda) - \lambda^T b}_{:=d(\lambda)}$$

With the subgradient of d given by:

$$\partial d(\lambda) = A \partial f^*(-A^T \lambda) - b$$

Recall $v \in \partial f^*(u) \Leftrightarrow u \in \partial f(v)$ which means that the optimizer in $-\sup_{x \in \mathbb{R}^n} \{(-\lambda^T A)x - f(x)\}$ satisfies:

$$-A^T \lambda \in \partial f(x^*) \Leftrightarrow x^* \in \partial f^*(-A^T \lambda)$$

As a Result, the subgradient $\partial d(\lambda)$ can be expressed via

$$\partial d(\lambda) = Ax - b, \text{ where } x \in \operatorname{argmin}_{\hat{x} \in \mathbb{R}^n} \{f(\hat{x}) + \hat{x}^T A^T \lambda\}$$

Dual Subgradient Method with step size $T_k > 0$:

$$x_k \in \operatorname{argmin}_{\hat{x} \in \mathbb{R}^n} \{f(\hat{x}) + \hat{x}^T A^T \lambda_k\}, \lambda_{k+1} = \lambda_k + T_k (Ax_k - b)$$

8.2 Example 1

Starting from (9) and with $Ax = 0$ s.t. $x_1 - x_2 = x_2 - x_3 = \dots = x_m - x_1 = 0$

BLACKBOARD

$$x_k \in \operatorname{argmin}_{x_1, \dots, x_m \in \mathbb{R}^n} \left(\sum_{i=1}^m f_i(x_i) \right) + \lambda_1(x_1 - x_2) + \lambda_2(x_2 - x_3) + \dots$$

With that the subgradient becomes

$$x_{k_i} \in \operatorname{argmin}_{\hat{x}_i \in \mathbb{R}^n} \{f_i(\hat{x}_i) - \lambda_{k_{i-1}}^T \hat{x}_i + \lambda_{k_i}^T \hat{x}_i\}$$

for $i = 2, 3, \dots, m-1$ in parallel

$$\lambda_{k+1, i} = \lambda_{k, i} + T_k (x_{k_i} - x_{k_{i+1}})$$

8.3 Real life examples

Video Quadcopter

- Not attached Pendulum

- Nonconvex OP

- Trajectory offline computed

- Track it with time-varying LQR feedback controller

Video Robotarm

- Table tennis

- Very flexibel arm

Dynamic control of magnetic navigation

- Balance stick on 4 magnets

- Precise control of fields

8.4 Example 2

$$f(x) = \sum_{i=1}^m f_i(x_i) \text{ with } Ax = b$$

$$x = (x_1, \dots, x_n) \text{ and } A = [A_1, \dots, A_m]$$

Dual subgradient becomes

$$x_{k_i} \in \operatorname{argmin}_{\hat{x}_i} \{f_i(\hat{x}_i) + \lambda_k^T A_i \hat{x}_i\} \text{ (local minimization)}$$

$$\lambda_{k+1} = \lambda_k + T_k (\sum_{i=1}^m A_i x_{k_i} - b) \text{ (broadcasting)}$$

IMAGE

Proposition 13. f convex with closed epigraph, f is μ -strongly convex if and only if f^* is $1/\mu$ -smooth.

From that we conclude

$$d(\lambda) = -f^*(-A^T \lambda) - \lambda^T b$$

f μ -strongly convex $\rightarrow f^*$ is $1/\mu$ smooth $\rightarrow d(\lambda)$ is $\bar{\sigma}(AA^T) 1/\mu$ -smooth

f is L -smooth $\rightarrow f^*$ is $1/L$ strongly convex $\rightarrow d(\lambda)$ is $\bar{\sigma}(AA^T) 1/L$ -smoothly convex

Problem f μ -strongly convex is hardly restricting condition

8.5 ADMM

$$\min_{x \in \mathbb{R}^n} f(x) + \frac{\rho}{2} |Ax - b|^2 \text{ s.t. } Ax = b \text{ with } \rho > 0$$

Augmented Lagrangian

$$x_k = \operatorname{argmin}_{x \in \mathbb{R}^n} f(x) + \lambda_k^T Ax + \frac{\rho}{2} |Ax - b|^2$$

$$\lambda_{k+1} = \lambda_k + T_k (Ax_k - b) \text{ (typically } T_k = \rho)$$

ADVANTAGE

Much better convergence properties even if f non-strongly convex.

DISADVANTAGE

Loose of decomposability / parallelization due to augmentation with quadratic term.

This motivates ADMM which tries to combine the best of both worlds. (Well conditioned minimization and parallelization)

8.6 Alternating direction method of multipliers

CONSIDER f,g

form augmented objective

augmented Lagrangian

ADMM

$$x_k = \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \mathcal{L}_p(x, z_{k-1}, \lambda_k)$$

$$z_k = \underset{z \in \mathbb{R}^n}{\operatorname{argmin}} \mathcal{L}_p(x_k, z, \lambda_k)$$

$$\lambda_{k+1} = \lambda_k + \rho(Ax_k + Bz_k - c)$$

EXAMPLE Images Low/High rank

8.7 Recitation

Recap

Optimization Problem

min f,g

Augmented Lagrangian

ADMM

$$x_k = \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \mathcal{L}_p(x, z_{k-1}, \lambda_k)$$

$$z_k = \underset{z \in \mathbb{R}^n}{\operatorname{argmin}} \mathcal{L}_p(x_k, z, \lambda_k)$$

$$\lambda_{k+1} = \lambda_k + \rho(Ax_k + Bz_k - c)$$

We can also consider completing the square in the augmented Lagrangian as

$$\mathcal{L}_p(x, z, \lambda) = f(x) + g(z) + \frac{\rho}{2} |Ax + Bz - c + \frac{\lambda}{\rho}|^2 - \frac{1}{2\rho} |\lambda|^2$$

and introduce new dual variable $\mu = \frac{\lambda}{\rho}$ to obtain a scaled version of ADMM.

SYSTEM

x_k

z_k

ν_{k+1}

Constrained optimization via ADMM

min...

Solve with ADMM:

1. Transform

2. Apply ADMM

Solving QPs with ADMM

QP

1. Transform to ADMM form:

min f,g s.t. A..

2. Apply ADMM

x_k

z_k

ν_{k+1}

3. Simplify the minimization steps

x-minimization is again QP with constraints

- Lagrangian

- KKT $\nabla_x \mathcal{L}(x, \mu) = \dots = 0 \Leftrightarrow$ Matrix system

z-minimization

- since g indicator function ...

- ..projection step..

$$z_k = \underset{\mathbb{R}_+^n}{\operatorname{prox}}(x_k + \mu_k)$$

9 Distributed optimization with ADMM

Motivation

- Slides - Distributed computation - Vanilla vs averaging

We start with:

$$\min_{x \in \mathbb{R}^n} \sum_{i=1}^m f_i(x)$$

Goal Solve problem such that each term can be handled by its own processor.

Reformulation:

$$\min_{x_1, \dots, x_N \in \mathbb{R}^n, z \in \mathbb{R}^n} \sum_{i=1}^N f_i(x_i) \quad \text{s.t.} \quad x_i = z, \quad i = 1, \dots, N \quad (10)$$

and apply ADMM

9.1 Global consensus problem

Solve (10) with ADMM

Step 1: Form augmented Lagrangian

$$\mathcal{L}_p() = SUM$$

Step 2: Formulate ADMM

$$\lambda_i^{k+1} = \lambda_i^k + \rho(x_i^{k+1} - z^{k+1})$$

$$\rho N z_{k+1} = \dots$$

$$\sum_{i=1}^N \lambda_i = \sum_{i=1}^N \{\lambda_i^k - \lambda_i^k\} = 0$$

therefore with $\lambda_i = 0$ for $i = 1, \dots, N$

$$z^{k+1} = \dots$$

this results in...

GRAFIK

9.2 Sharing Problem

$$\min_{x_1, \dots, x_N \in \mathbb{R}^n} \sum_{i=1}^N f_i(x_i) + g\left(\sum_{i=1}^N x_i\right)$$

Apply ADMM:

$$x_i = z_i, i = 1, \dots, N$$

Step 1: Form augmented Lagrangian

$$\mathcal{L}_p(x_1, \dots, x_N, z_1, \dots, z_N, \lambda_1, \dots, \lambda_N) = \sum_{i=1}^N f_i(x_i) + g\left(\sum_{i=1}^N z_i\right) + \rho \dots$$

Step 2: Formulate ADMM dynamics

$$x_i^{k+1} \quad (11)$$

$$z_i^{k+1} \quad (12)$$

$$\lambda_i^{k+1} \quad (13)$$

Simplify (12) with $a_i = \dots$

stationary conditions for (12) $O \in$

...greatly simplified by introducing averages $\bar{z}^{k+1}, \bar{a}$

Then we arrive at N stationary conditions...

R

NR

O

$$\bar{z}^{k+1}$$

$$z^{k+1}$$

$$\lambda_i^{k+1}$$

all λ_i^{k+1} equal

FINAL DYNAMICS

$$x_i^{k+1}$$

$$\bar{z}^{k+1}$$

$$\lambda^{k+1}$$

Principles: (not shown)

Dual of Sharing Problem

derivations (not shown)

sup

9.3 Optimization over Graphs

$g = (V, E)$ undirected graph with vertices V and edges E

Solve

$$\min_{x \in \mathbb{R}^n} \sum_{i \in V} f_i(x)$$

where each vertex has local data and we would like to fit a model with shared parameters

GRAFIK

Idea Reformulation with constraints

$$\min_{x_1, \dots, x_{|V|}, z_1, \dots, z_{|V|}} \sum_{i \in V} f_i(x_i) \text{ s.t. } x_i = z_{ij}, x_j = z_{ij} \quad \forall (i, j) \in E$$

Step 1: Form augmented Lagrangian

$$\mathcal{L}_p() = SUM + SUMSUM$$

Step 2: Formulate ADMM

DERIVATIONS

FINAL RESULTS

$$x_i^{k+1} \in \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} f_i(x_i) + SUM$$

$$\bar{x}_i^{k+1}$$

$$p_i^{k+1}$$

9.4 Recitation

QUIZ Questions

2023 1b) conjugate function

2020 2a) Hyperplane with dual-minimization

10 Signal denoising and regression

Linear equation $y = Ax, y \in \mathbb{R}^n, x \in \mathbb{R}^m, A \in \mathbb{R}^{m \times n}$

- classic setting $m \gg n$

- modern setting $m \ll n$

10.1 Classic setting with outliers

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2$$

uses l_2 -norm to penalize large residuals

GRAFIK

but as a result, outliers have a lot of weight

Grafik

Weight of outliers can be reduced with l_1 -norm

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2 \quad (14)$$

Rewrite (14) as convex program

min,sum,zi

s.t.

AGAIN REformulate

-> linear program

For best of both worlds:

$$\phi_{\text{Hub}}(u) = \begin{cases} u^2 & \text{if } |u| \leq M \\ 2Mu - M^2 & \text{if } |u| > M \end{cases}$$

resulting OP:

min,sum,fub,()

10.2 Modern setting

- $Ax = y$ has infinitely many solutions

- Which one is the best?

- add regularizer

Tikhonov regularizer: $\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2 + \lambda \|x\|_2^2$

Least Absolute Shrinkage and Selection Operator:

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2 + \lambda \|x\|_1$$

is equivalent to

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2 \text{ s.t. } \|x\|_1 \leq c$$

Example

Audio signal, $f_1 = 102$ Hz, $f_2 = 305$ Hz

$$\tilde{x}(t) = \cos(2\pi f_1 t) + \cos(2\pi f_2 t) + n(t) \quad (15)$$

Signal evaluated at 100 randomly selected points $t_i \in [0, 1]$

MATLAB script

-> how choose λ ?

Projection on l_1 Ball

Approach to solve $\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|_2^2 \text{ s.t. } \|x\|_1 \leq c$ with projection:

$$\operatorname{argmin}_{x \in \mathbb{R}^n} \frac{1}{2} \|x - y\|_2^2$$

results in Lagrange function:

$$\begin{aligned} \mathcal{L}(x, \lambda) &= \frac{1}{2} \|x - y\|_2^2 - \lambda(c - \|x\|_1) \\ &= \left(\sum_{i=1}^n (x_i - y_i)^2 + \lambda \|x\|_1 \right) - \lambda c, \lambda \geq 0 \end{aligned}$$

where we set $l_i(x, \lambda) = \frac{1}{2} (x_i - y_i)^2 + \lambda |x_i|$

Figure of $\partial_x l_i$ with respect to x_i

IMAGE

Result:

$$\begin{aligned} x_i &= \left\{ \operatorname{argmin}_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda) \right\}_i = \{t_1\} \\ &= \{t_1\} \\ &= \{t_1\} \end{aligned}$$

-> how choose λ ?

if ... then ..

Example 3 Image denoising

Example 4 Face recognition

10.3 Recitation

11 Classification

$\tilde{y}(\tilde{x})$ takes values in discrete categories

Setup: dataset of $(\tilde{x}_i, \tilde{y}_i)$, $i = 1, \dots, N$ with $\tilde{x}_i \in \mathbb{R}^n$, $\tilde{y}_i \in \{1, 2, \dots, K\}$

Naive Approach

Classify with

$$f^{\text{naive}}(\tilde{x}) =$$

11.1 Recitation

12 Adaptive decision-making

12.1 Recitation